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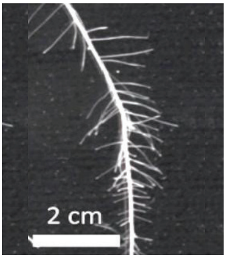
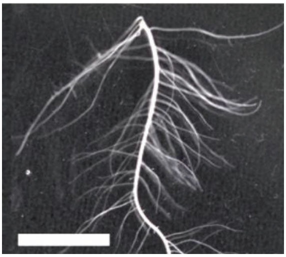
ENHANCED ROOT SYSTEM DEVELOPMENT UNDER VARIOUS PHOSPHORUS CONDITIONS THROUGH EFFICIENT CARBOHYDRATE ALLOCATION

INTRODUCTION

Roots play a vital role in plant growth. They serve as anchors, absorb water and nutrients, and support above-ground growth. Studies have shown that enhanced root systems can improve plant's resilience to environmental stresses, such as drought and nutrient deficiency. By developing root systems that are more efficient in extracting resources from the soil, plants can better withstand adverse conditions and still produce high yields. By comparing the wild-type Nipponbare, a japonica rice cultivar, and the 11NB10, a new mutant strain of rice, the role of carbohydrates in regulating root growth and development under different phosphorus (P) conditions was identified.



Wild-type Nipponbare
(Normal gene)



11NB10 mutant
(Mutated gene)

KEY RESULTS

- The 11NB10 mutant showed promoted root growth under low P application.
- Root growth of 11NB10 was further enhanced when P application was increased.
- 11NB10 mutant showed lower starch content at the basal shoot at P-deficient condition.
- Starch content at the basal shoot of 11NB10 was further reduced when P application was increased.
- Glucose contents in the root regions of 11NB10 were higher than the wild-type Nipponbare.

IMPLICATION

- 11NB10 mutant has a significant advantage over the wild-type Nipponbare in terms of acquisition of nutrients, especially phosphorus.
- 11NB10 mutant's ability to efficiently allocate carbohydrates to root growth is a key factor contributing to its superior performance.

POTENTIAL APPLICATION

- The enhanced root system of the 11NB10 mutant has a significant potential for improving rice yields, particularly in phosphorus-deficient soils, and mitigating climate change impacts.
- By identifying genes responsible for this trait of 11NB10 and incorporating them into breeding activities, researchers can develop rice varieties with superior root architecture, leading to increased nutrient uptake, better drought tolerance, and overall crop resilience.

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ABOUT THE MATERIAL

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